Applications of Artificial Intelligence in Structural Engineering

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Abstract

AI has the potential to revolutionize the way we design, build, and maintain our built environment. The paper provides an overview of various AI branches, including deep learning, neural networks, swarm optimization, fuzzy logic, machine learning, pattern recognition, probability theory, and metaheuristic optimization. It highlights the applications of AI in structural engineering, such as structural health monitoring, modular construction decision making, and vibration control. The paper also examines the limitations, challenges, and opportunities associated with the adoption of AI in the industry. The paper concludes that AI has enormous potential to enhance the efficiency, safety, and sustainability of structural engineering, and that further research and collaboration between academia and industry are necessary to fully realize the benefits of this technology.

Keywords— artificial intelligence, structural engineering, machine learning, pattern recognition, civil engineering.

I. Introduction

In the world today, Artificial Intelligence (A.I.) is constantly evolving and percolating into all faucets of life, the term encompasses all aspects in the field of science and engineering which deals with the use of computers to comprehend and replicate what is typically referred to as intelligent actions or conduct, as well as the development of objects that demonstrate such conduct.

The field dates back to the mid-20th century, when researchers defined AI at the Dartmouth Conference in 1956 [1]. AI research progressed rapidly in the following years, with the development of early AI systems such as the Logic Theorist and the General Problem Solver [2]. In the 1960s and 1970s, AI research faced setbacks during the "AI winter," but it experienced a resurgence in the mid-1980s due to advances in computing and algorithm development [3]. In the 21st century, AI has become an increasingly important field, with applications in a wide range of industries, and ongoing research in areas such as deep learning, reinforcement learning, and explainable AI [4]. The history of AI and be categorized into five distinct eras: incubation period before 1956, formation period from 1956 to 1970, dark period from 1966 to 1974, knowledge period from 197- to 1988, and integrated development period from 1986 till date [5]. This has seen AI become capable of natural language processing, autonomous vehicle driving, data analysis, computer visualization and design, and so on [6]. In spite of the feats attained, AI is still capable of much more, the application of AI is ubiquitous and although its usage is evident in many fields, this paper focuses on its application in structural engineering.

Structural engineering is a critical aspect of building design and construction, with the ability to ensure that structures are safe, efficient, and sustainable. Recent advancements in Artificial Intelligence (AI) have opened up new opportunities to enhance the field of structural engineering, with the potential to improve safety, reduce costs, and optimize designs. The possibilities of AI in structural engineering are limitless, with current usage in processing large amounts of data, identifying patterns, and predicting structural behavior, while providing engineers with valuable insights so as to make more informed decisions. The application of AI in structural engineering has already begun to revolutionize the field, with examples ranging from the design of high-rise buildings to the monitoring of infrastructure. However, there are also challenges and limitations that need to be addressed, such as the need for large amounts of data and the potential for errors in AI algorithms. In this paper, we will explore the current state of the art in AI applications in structural engineering, highlighting recent advances, challenges, and future directions.

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AI usage in structural engineering has given rise to designated branches:

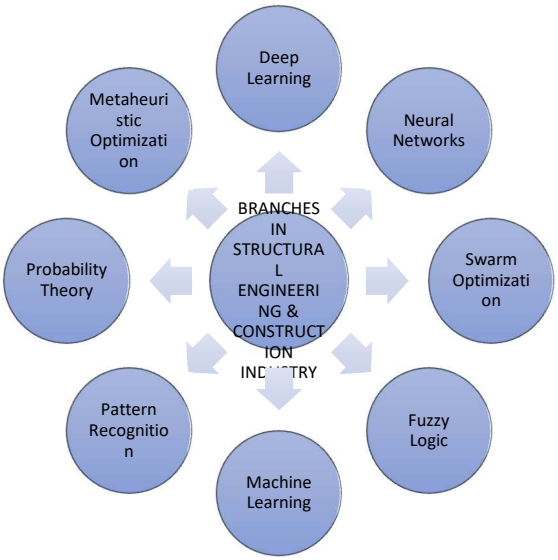

Figure 1: AI branches in structural engineering & construction

As seen in Figure 1 above, the continual evolution of AI in the field of structural engineering is evident in how many sub-disciplines that have emerged, with each branch offering unique benefits. Deep Learning, Neural Networks, and Pattern Recognition allow for the automatic recognition of patterns, classification, and prediction of structural behavior [7]. Swarm Optimization, Metaheuristic Optimization, and Probability Theory offer optimization of design and construction processes, while Fuzzy Logic provides a framework for reasoning with uncertain information. Machine Learning enables the automatic learning of rules and decision-making based on past data, enabling more informed and accurate decisions [8]. By combining these AI branches, structural engineers can enhance the safety, efficiency, and sustainability of their designs and construction projects.

II. APPLICATIONS OF AI IN STRUCTURAL ENGINEERING

As stated earlier, AI has various application in the field of structural engineering at pre-construction (Planning), construction and post-construction phase (maintenance).

Design optimization: this is currently the most used application of AI in structural engineering, companies like Cazza, Doxel, and Buildots already use AI for design optimization. AI algorithms can be used to optimize structural designs, taking into account various constraints and objectives, such as strength, safety, and cost [9]. These algorithms can quickly explore many design options and select the most optimal one [10]. This can lead to more efficient and cost-effective designs. This is progressive as machine learning algorithms can learn from past designs and suggest modifications to improve the performance of new designs.

Structural analysis: AI can be used to analyse complex structural systems, predict structural behaviour, and identify potential failures [11]. Deep learning algorithms can recognize patterns in large data sets and make predictions about the behaviour of structures under different loads and conditions [12]. This can help engineers to identify potential problems early on and take preventive measures.

Monitoring and maintenance: the branch called machine learning is used to monitor the structural health of buildings and infrastructure. Sensors can be installed in a bridge to detect changes in its behaviour and alert engineers to potential problems [13]. AI algorithms can also be used to schedule maintenance tasks and optimize maintenance schedules. This can help to prolong the lifespan of structures and reduce maintenance costs.

Construction planning: AI algorithms can be used to optimize construction schedules and resource allocation, taking into account various constraints and objectives. This can help to reduce construction time and costs. Also, software such as Alice is being used to find optimal plans which could be further tested using an AI tool called Stack [14]. Machine learning algorithms can predict the duration of different construction activities and help to schedule them in the most efficient way.

Quality assurance and control: the quality of a structure is critical; AI can be used to monitor and ensure the quality of construction materials and processes. For example, computer vision algorithms can detect defects in concrete or steel. This can help to prevent failures due to substandard materials or workmanship. After defects have been identified, trial-and-error method can be adopted by the AI to propose Good International Industry Practice (GIIP) [15]. 3D modelling has evolved from just design visualization to becoming a tool for quality assurance as completed building models are benchmarked by initial models to ascertain building defects and deviation from initial planning.

Safety and Risk Control: AI can be used to improve the safety of construction sites by identifying potential hazards and alerting workers to potential dangers. Machine learning algorithms can analyse video feeds from construction sites and identify unsafe practices or conditions [16].

Artificial Neural Network (ANN) can be used by engineers for predictions and preventions of failures, ensuring building risk mitigation.

A. Emerging usage of AI in Structural Engineering

1) Structural Health Monitoring (SHM)

Structural health monitoring (SHM) is the process of using sensors and other technologies to continuously or periodically monitor the condition and performance of a structure over its lifetime [7]. The aim of SHM is to detect and locate any damage or deterioration that may occur, as well as to assess the remaining useful life of the structure [8].

There are two types of structural health monitoring (SHM): one that uses computer vision technology to analyze images (image-based) and the other that uses

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signals obtained during dynamic events to monitor vibrations (signal-based) [9].

SHM can be applied to a wide range of structures, including buildings, bridges, tunnels, and offshore platforms [10].

Some of the key benefits of SHM include:

Early detection of damage: By continuously monitoring the structure, SHM can detect and locate damage or deterioration at an early stage, allowing for timely repair and maintenance [11].

Improved safety: SHM can help to improve the safety of structures by identifying potential hazards and preventing catastrophic failures [12].

Cost savings: By detecting damage early, SHM can help to reduce repair costs and extend the lifespan of structures [13].

Enhanced performance: SHM can help to optimize the performance of structures by providing real-time data on their behavior and identifying areas for improvement.

Some of the technologies commonly used in SHM include sensors for measuring strain, temperature, vibration, and other parameters, as well as data acquisition systems for collecting and processing sensor data. AI and machine learning algorithms can also be used to analyze sensor data and make predictions about the behavior of the structure [14]. Convolutional Neural Networks (CNNs) and other deep learning algorithms have been used to automatically extract features [15]. With these methods, the feature extraction and classification steps are combined into one step to avoid the need for exhaustive tests between features and classifiers through trainings.

Although machine learning (ML) and deep learning (DL) are subfields of artificial intelligence (AI) they differ in their approach to learning from data. ML algorithms learn patterns in data through statistical analysis and are useful for a range of applications, while DL algorithms learn multiple layers of representations in data using neural networks and are particularly suited for complex tasks like natural language processing and computer vision. This is illustrated in Figure 2 below:

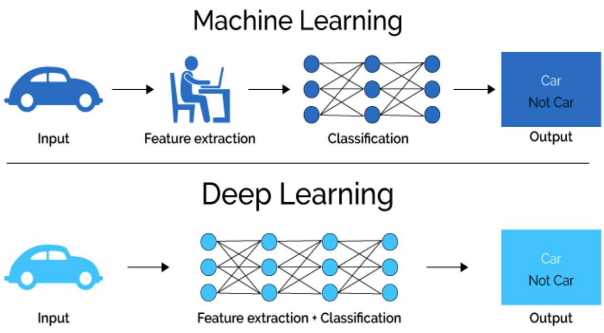

Figure 2: Machine Learning and Deep Learning processes

2) Structural System Identification

Structural system identification is the process of determining the dynamic properties of a structure using measured data [7]. This involves analyzing the response of the structure to known or unknown forces, and then using mathematical models to identify the system properties such as natural frequencies, damping ratios, and mode shapes [8]. System identification is an important tool in structural health monitoring, which is used to detect and diagnose structural damage, assess the remaining useful life of structures, and make informed decisions about maintenance and repairs [9]. Accurate and reliable system identification requires a combination of experimental measurements, data analysis techniques, and mathematical modeling, and it is a key area of research in the field of structural engineering [10].

3) Construction Management System

A Construction Management System (CMS) is a software application that is designed to help construction managers and contractors streamline their workflows and manage various aspects of construction projects. A CMS typically includes modules for project planning, scheduling, resource management, budgeting, and documentation. It allows users to track progress, monitor costs, and manage communications and collaboration among project team members [29].

CMS systems may also include features for Building Information Modeling (BIM), which is a digital representation of a building or structure that can be used for design, construction, and maintenance purposes. BIM models can be used to visualize the project and identify potential conflicts and issues before construction begins [20].

CMS can provide many benefits, including increased efficiency, better project management, and improved communication and collaboration among project team members. The use of CMS has become increasingly popular in recent years as construction projects become more complex and larger in scale [30].

4) Modular Construction Decision Making

Modular construction decision making refers to the process of evaluating the feasibility and benefits of modular construction compared to traditional on-site construction. Modular construction involves assembling building components off-site in a factory or other controlled environment, and then transporting them to the construction site for final assembly. This approach offers several advantages over traditional on-site construction, including faster construction times, reduced waste, and improved quality control [31].

The decision to use modular construction involves evaluating factors such as project size, complexity, location, schedule, and budget. Other considerations include the availability of skilled labor, local building codes and regulations, and transportation logistics [14].

To make informed [13] decisions about modular construction, project teams may use a variety of tools and techniques, such as cost-benefit analysis, risk assessment, and value engineering [32]. These methods can help identify the potential benefits and challenges of modular construction, and inform the decision-making process.

5) Neural Dynamic Classification Algorithm

Neural dynamic classification algorithm is a machine learning technique used for classification tasks. It is based on the concept of neural networks, which are modeled after

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the structure and function of the human brain. The algorithm is designed to learn and classify patterns in data that are not easily identified by traditional statistical methods [18].

The neural dynamic classification algorithm consists of a network of interconnected nodes that process and classify input data. During the training phase, the algorithm adjusts the weights and connections between the nodes to optimize the classification accuracy [28]. This process continues until the algorithm can accurately classify the input data.

The neural dynamic classification algorithm has several advantages over traditional classification methods. For instance, it can handle complex and high-dimensional data, adapt to changes in the data over time, and make decisions in real-time. It is particularly useful for classification tasks in dynamic environments where the data is constantly changing, such as in signal processing, image recognition, and natural language processing.

6) Vibration Control

Vibration control of structures is an important area of study in structural engineering as excessive vibrations can cause damage to buildings and other structures. Artificial intelligence (AI) techniques, such as machine learning and neural networks, have been used to develop advanced control strategies for reducing vibrations in structures [7].

One common approach is to use AI-based control systems to optimize the damping properties of structures, which can reduce the effects of vibrations caused by external forces such as earthquakes or wind loads. These systems use sensors to monitor the vibration levels of the structure in real-time and adjust the damping properties accordingly to minimize the vibrations [8].

Another approach is to use AI-based algorithms to develop active vibration control systems, which use actuators to apply forces to the structure in order to reduce the vibration levels. These systems can be designed to adapt to changing environmental conditions and can provide more effective vibration control than traditional passive damping systems [9].

AI-based techniques have also been used for damage detection and health monitoring of structures, which can help to prevent excessive vibrations by identifying and repairing structural damage before it becomes a problem. Machine learning algorithms can analyze data from sensors and detect changes in the vibration patterns of structures, which can indicate the presence of damage or defects [10].

B. Limitations

Although AI has shown great potential in revolutionizing the field of structural engineering, there are still some limitations to its applications. One of the main limitations is the lack of reliable and comprehensive data for training AI algorithms. Without sufficient data, AI models may not be accurate or may fail to generalize to new situations [36]. Another limitation is the potential for bias in AI models, which can lead to incorrect predictions or decisions. Additionally, the complexity and cost of implementing AI systems may be a barrier to their widespread adoption in the industry [37]. Finally, there is the concern about the impact of AI on employment, as AI systems may replace human workers in certain tasks. Therefore, it is important to carefully consider the limitations and potential risks of AI applications in structural engineering.

C. Challenges

The applications of AI in structural engineering also face several challenges. One of the major challenges is the need for domain expertise in both AI and structural engineering. It can be difficult to find individuals who possess both sets of skills and knowledge [11]. Another challenge is the lack of standardization in the field, which can make it difficult to compare and evaluate different AI algorithms [31]. Additionally, the large amounts of data required for training AI models can be challenging to acquire and process, especially for smaller organizations. There is also the challenge of integrating AI systems with existing software and hardware in the industry. Finally, there is a need for further research and development to improve the accuracy and reliability of AI algorithms in complex structural systems [22]. Overcoming these challenges will require collaboration between experts in AI and structural engineering, and continued investment in research and development.

D. Opportunities

The applications of AI in structural engineering offer numerous opportunities for the industry. One of the key opportunities is the potential to improve the efficiency and accuracy of design and analysis processes [3]. AI algorithms can be used to quickly process large amounts of data, identify patterns and relationships, and generate optimized design solutions [2]. This can lead to significant time and cost savings for structural engineering firms.

Another opportunity is the ability of AI systems to improve safety and reliability in structural systems. By continuously monitoring and analyzing structural data, AI systems can detect potential failures or abnormalities in real-time, allowing for timely maintenance and repairs [22]. This can help prevent catastrophic failures and ensure the safety of people and property.

AI systems can also enable the use of more advanced and complex structural designs, as they can process and analyze complex data sets that would be difficult for humans to interpret [24]. This could lead to the development of more innovative and sustainable building designs that are optimized for their specific environments.

III. Conclusion

In conclusion, AI has emerged as a promising tool for solving various complex problems in the field of structural engineering. It has the potential to improve structural design, optimize construction processes, enhance structural health monitoring, and enable more efficient and effective maintenance of infrastructure. The applications of AI in structural engineering are numerous, ranging from structural systems identification and vibration control to modular construction decision making and construction management systems. However, the adoption of AI in the industry also poses several challenges, such as data availability, lack of standardization, and the need for specialized expertise. To fully realize the potential of AI, these challenges need to be addressed through collaborations between academia, industry, and policymakers. Overall, the opportunities that AI presents for structural engineering are vast, and its integration into the field has the potential to transform the way we design, build, and maintain our built environment.

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